

Conjugated Systems

Organic Chemistry II

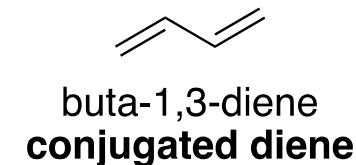
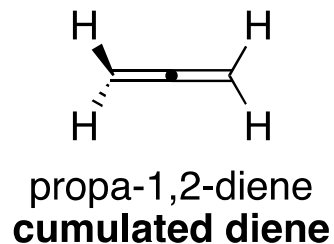
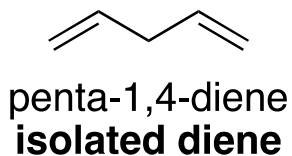
CHEM 2312

Dr. Michael Evans

1. Recognize and draw resonance forms of conjugated dienes, allylic cations, anions, and radicals, and other conjugated (delocalized pi) systems.
2. Describe and apply the conformational dynamics of conjugated dienes.
3. Interpret the pi molecular orbitals of conjugated systems.
4. Recognize electron-donating groups and electron-withdrawing groups and apply them in resonance.
5. Describe and apply the effects of EDGs and EWGs on frontier orbital energies to determine the relative strengths of conjugated molecules as nucleophiles or electrophiles.
6. Apply the HOMO-LUMO energy gap to explain the emergence of color in long conjugated polyenes.

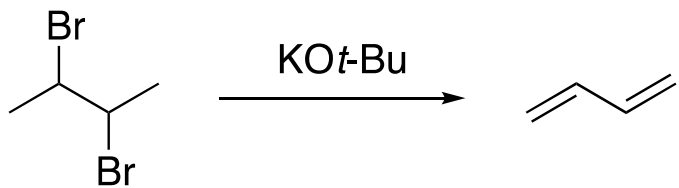
16.1 Isolated, Cumulated, and Conjugated Dienes

- **Isolated dienes** contain two alkene groups separated by two or more sigma bonds
- **Cumulated dienes** contain two carbon-carbon double bonds sharing a carbon (adjacent)
- **Conjugated dienes** contain two carbon-carbon double bonds separated by one sigma bond

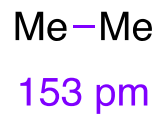
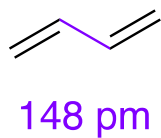


16.2 Preparation and Structure of Conjugated Dienes

- Conjugated dienes are readily prepared using elimination reactions
- Due to their special stability, conjugated systems often form preferentially when other possibilities exist

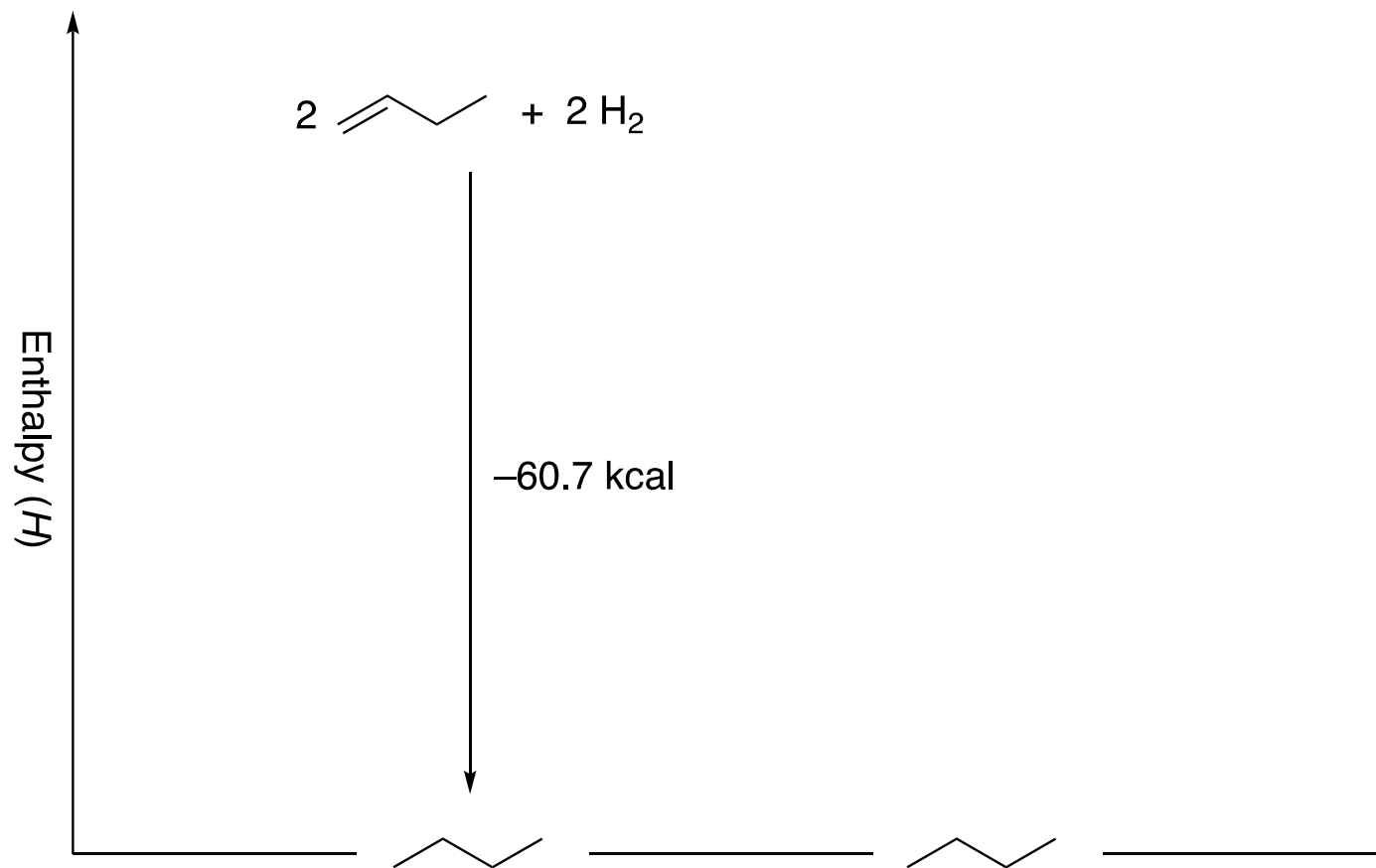


- The C–C single bond in conjugated dienes is shorter than a typical $\text{C}(sp^3)\text{--C}(sp^3)$ single bond



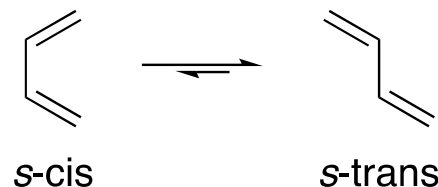
16.2 Conjugated Dienes are More Stable than Isolated Dienes

- Heats of hydrogenation show that conjugated dienes are more stable than isolated dienes

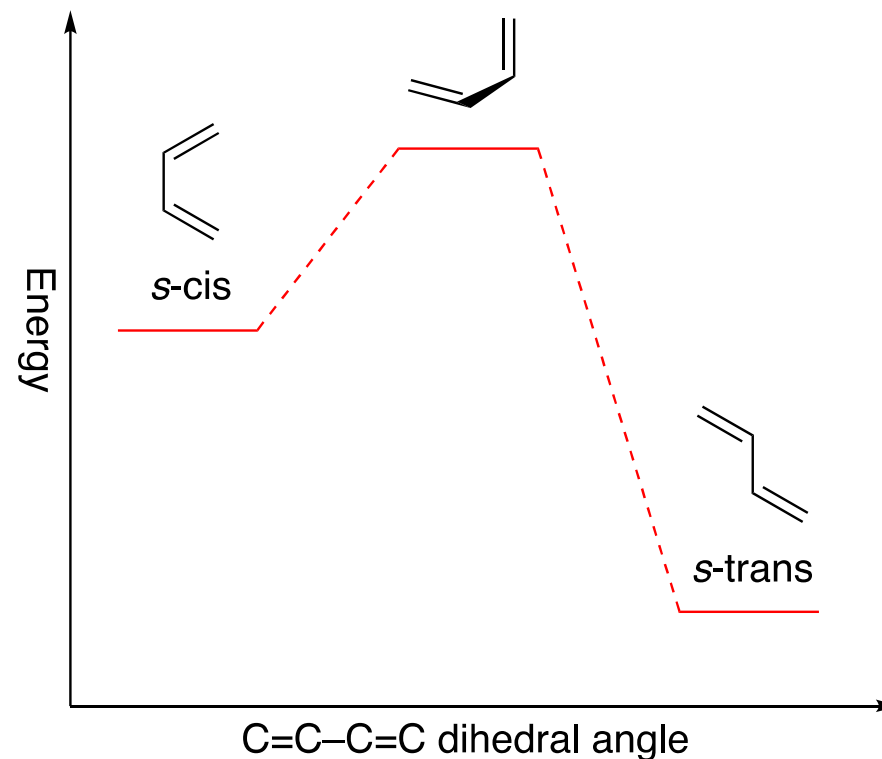


16.2 Rotation About the C–C Single Bond in Conjugated Dienes

- Conjugated dienes have two important conformations differing in rotation about the central carbon–carbon single bond: **s-cis** and **s-trans**



- Activation barrier for interconversion is 3.7 kcal/mol
- The s-trans conformer is more stable than the s-cis by 2.9 kcal/mol (98:2 at equilibrium)



- The pi molecular orbitals of conjugated systems are constructed from parallel atomic $2p$ orbitals
- We will use a computer program to calculate these ([HuLiS](#)) and focus on the **frontier orbitals**, the **highest occupied molecular orbital (HOMO)** and **lowest unoccupied molecular orbital (LUMO)**
- With increasing length of the conjugated system, the HOMO–LUMO energy gap shrinks

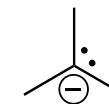
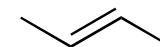
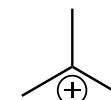
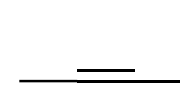
- The pi molecular orbitals of conjugated systems are constructed from parallel atomic $2p$ orbitals
- We will use a computer program to calculate these ([HuLiS](#)) and focus on the **frontier orbitals**, the **highest occupied molecular orbital (HOMO)** and **lowest unoccupied molecular orbital (LUMO)**
- For polyenes with an even number of carbon atoms, the HOMO resembles the dominant Lewis structure
- The LUMO resembles an α,ω -diradical resonance structure

- Any atom bearing an unhybridized p orbital can participate in conjugation, including heteroatoms bearing lone pairs, carbanions, and carbocations

- The structural elements of conjugated systems:

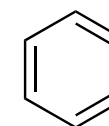
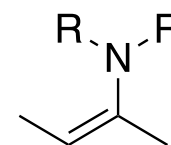
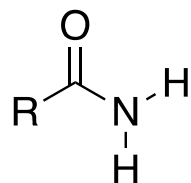
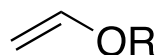
- Double and triple bonds (any atoms)
- Heteroatoms bearing a lone pair in a p orbital
- Carbon bearing a lone pair in a p orbital (carbanion)
- Trigonal carbocations and other six-electron atoms (e.g., group 13)

Structural elements of delocalized pi systems

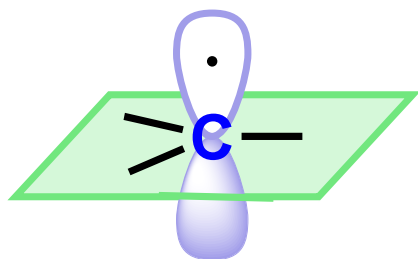
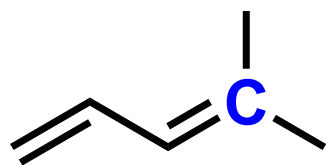


- At least one significant alternative resonance form can be drawn

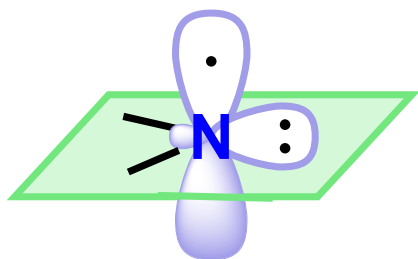
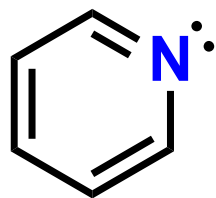
- Examples:



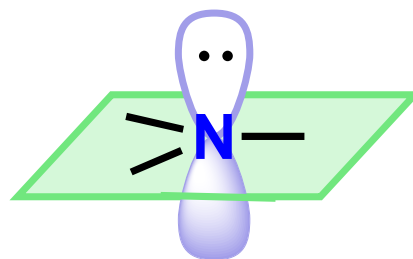
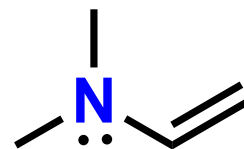
- Heteroatoms can contribute one and only one lone pair to a conjugated system, provided they are *not* involved in pi bonding



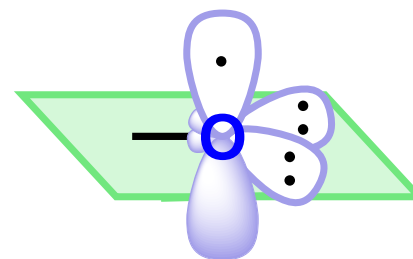
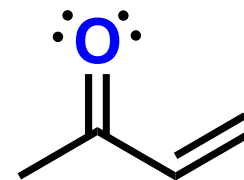
C



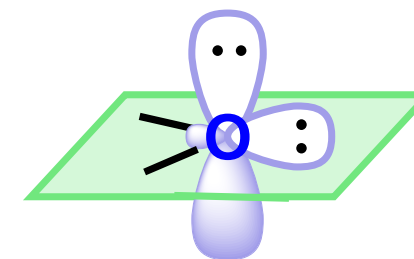
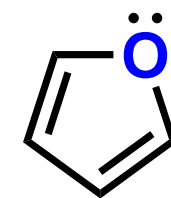
N2



N3

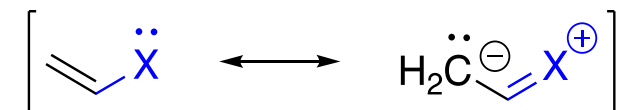


O1



O2

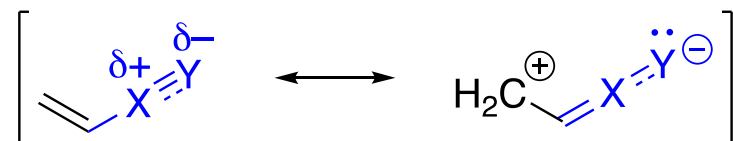
- **Electron-donating groups (EDGs)** raise the energy of the pi HOMO and “push” electron density to the carbons of a conjugated system



- In resonance forms, negative charge indicates carbons that are nucleophilic
- Examples: -OH, -OR, -NH₂, -NR₂, -SR

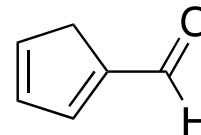
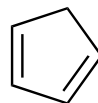
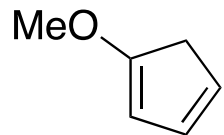
- **Electron-withdrawing groups (EWGs)** lower the energy of the pi LUMO and “pull” electron density from the carbons of a conjugated system

- Y is more electronegative than X

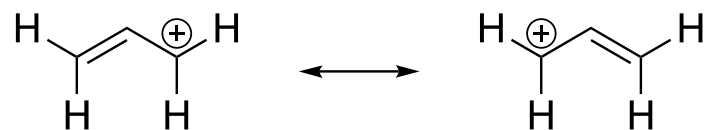


- In resonance forms, positive charge indicates carbons that are electrophilic
- Examples: $-\text{COR}$, $-\text{CHO}$, $-\text{CO}_2\text{R}$, $-\text{SO}_2\text{R}$, $-\text{CN}$, $-\text{NO}_2$

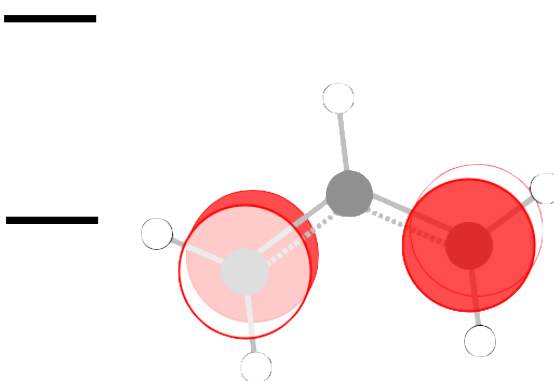
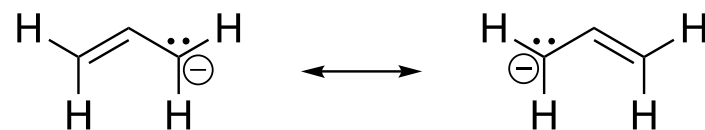
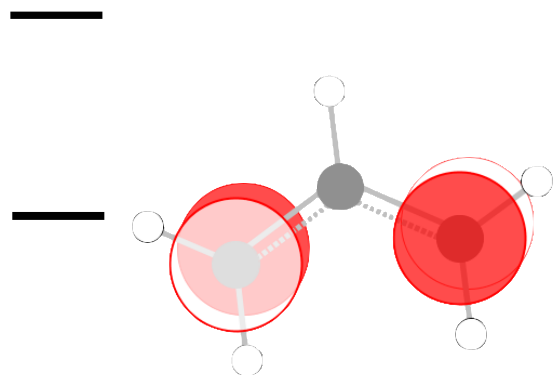
- Electrophilic addition involves a pi bond as the nucleophile. Which molecule would you expect to be most reactive in electrophilic addition with Br_2 ? Why?



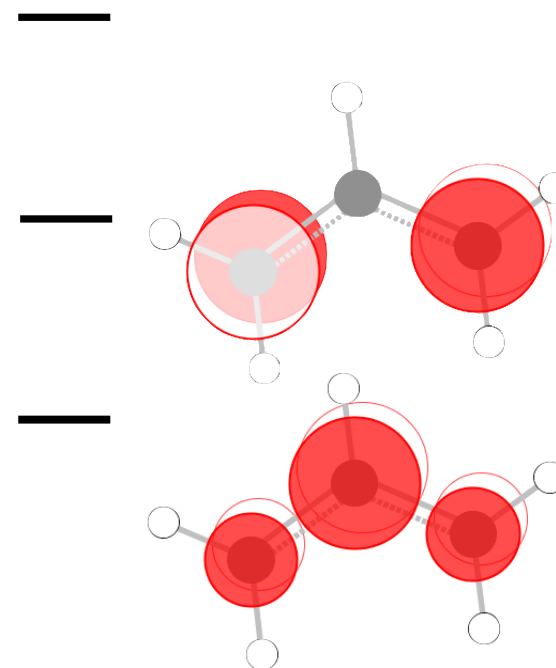
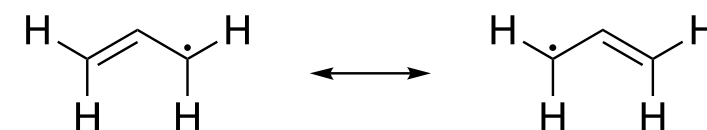
- The terminal atoms of the three-atom **allylic system** may have cationic, anionic, or radical character
- Allylic intermediates are more stable than comparable saturated counterparts



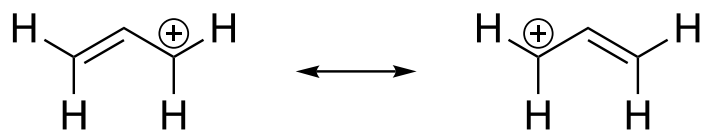
Positive charge shifts orbital energies down.



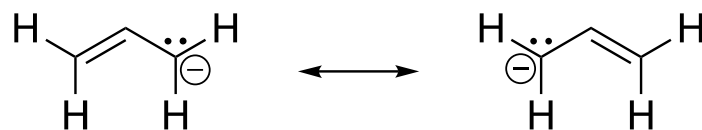
Negative charge shifts orbital energies up.



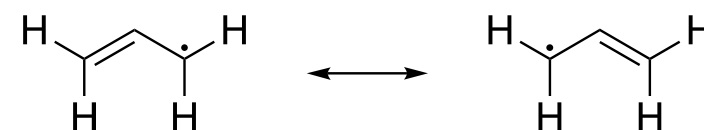
- Allylic reactive intermediates are relatively easy to access in reaction mechanisms
- However, they have multiple reactive sites...watch out for isomeric products!



Rapid S_N1 solvolysis of allylic (pseudo)halides



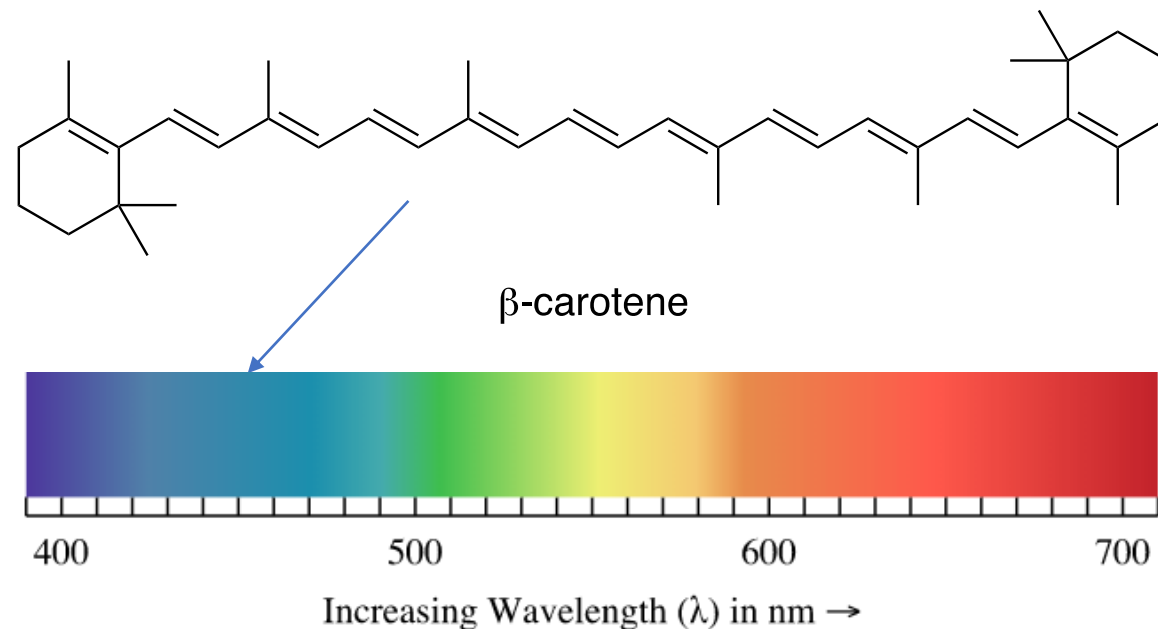
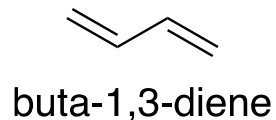
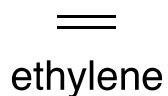
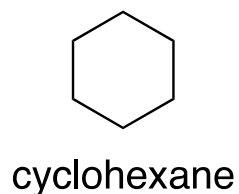
$pK_a(\text{H}_3\text{C}-\text{CH}=\text{CH}_2) = 43$



Allylic halogenation with NBS

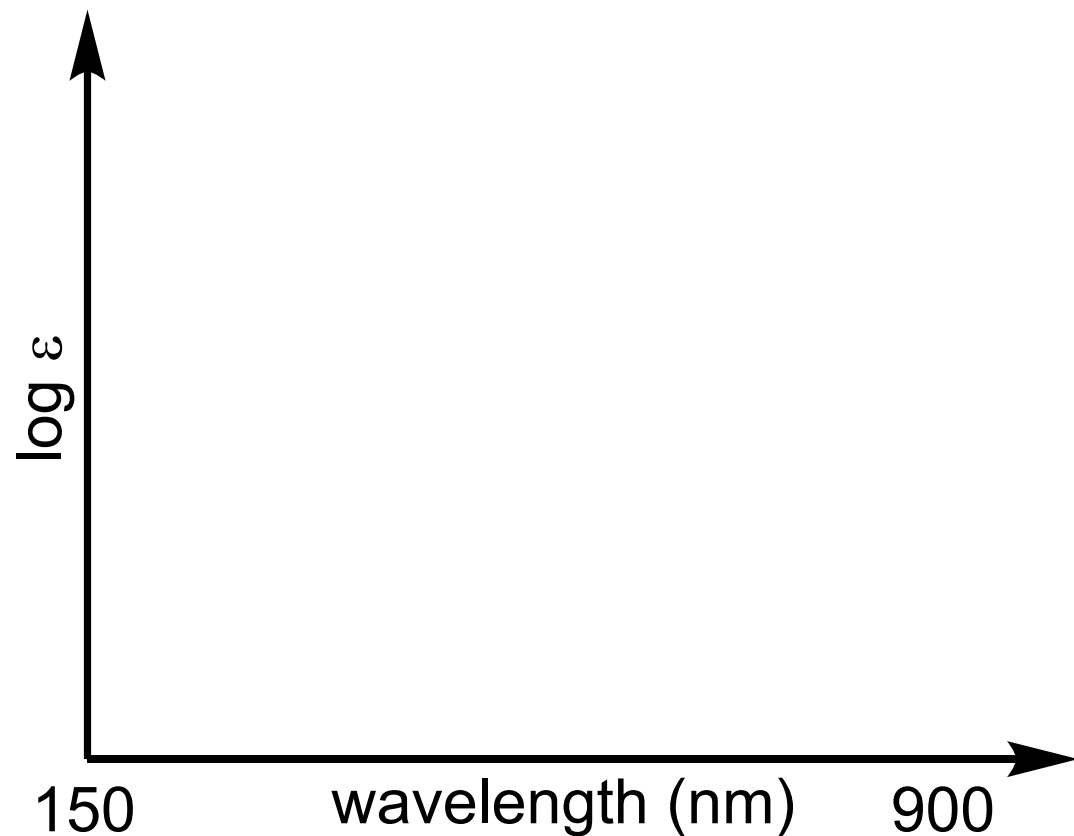
16.11 Molecular Origins of Color

- Compounds appear colored due to the absorption of visible light as electrons are excited from lower to higher (orbital) energy levels
- Most organic compounds absorb only in the ultraviolet *but* long conjugated systems can absorb in the visible range
- **Chromophores** are the groups within molecules that actually interact with and absorb/emit light
- The lowest-energy absorption typically involves promotion of an electron from HOMO to LUMO



16.11 UV-Vis Spectroscopy

- An **ultraviolet-visible (UV-vis) spectrum** shows the amount of light that a compound absorbs as a function of wavelength
- Each peak corresponds to excitation of an electron from a filled to an empty orbital
- The **wavelength of maximum absorbance** (λ_{max}) corresponds to the strongest transition



Compound	λ_{max} (nm)
$\text{H}_2\text{C}=\text{CH}_2$	165
$\text{H}_2\text{C}=\text{CH}-\text{CH}=\text{CH}_2$	217
$\text{H}_2\text{C}=\text{CH}-\text{CH}=\text{CH}-\text{CH}=\text{CH}_2$	258

Aromatic and Heteroaromatic Compounds

Organic Chemistry II

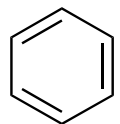
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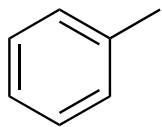
1. Recognize benzene rings and other aromatic structures within larger organic molecules.
2. Describe the relative positions of substituents linked to a benzene using *ortho/meta/para* notation.
3. Interpret the pi molecular orbitals of benzene and related aromatic hydrocarbons and aromatic heterocycles; apply the Frost circle to draw an orbital energy diagram for a fully conjugated cyclic hydrocarbon.
4. Apply the criteria for aromaticity to determine whether a given structure is aromatic, antiaromatic, or non-aromatic.
5. Draw resonance forms of aromatic heterocycles and apply them to recognize reactive sites in these compounds.
6. Engage benzylic carbons in reactions based on the relative stability of benzylic cations, anions, and radicals.
7. Apply resonance and inductive effects to recognize and describe electron density in substituted benzenes.

17.1 Introduction to Aromatic Compounds

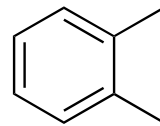
- Benzene and its derivatives belong to a class of compounds called **arenes (aromatic hydrocarbons)**



benzene

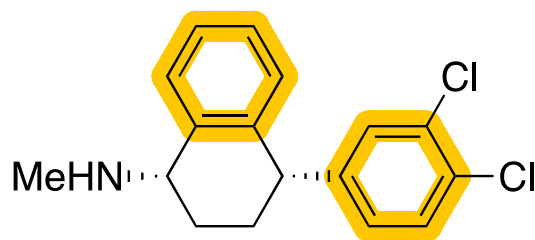


toluene

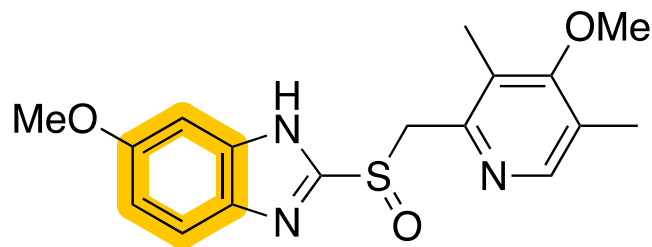


ortho-xylene

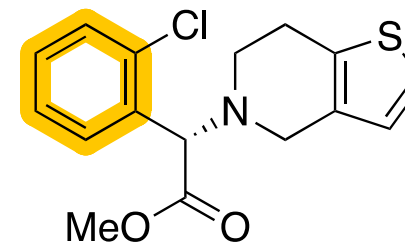
- Benzene rings are extremely common in organic compounds



Zoloft
(sertraline)



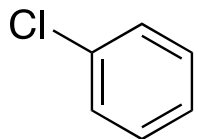
Prilosec
(omeprazole)



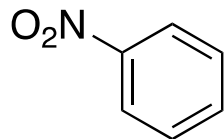
Plavix
(clopidogrel)

17.2 Monosubstituted Derivatives of Benzene

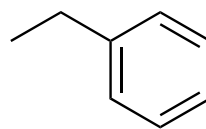
- Simple substituted benzenes are named using **benzene** as the parent name
- In larger molecules, a benzene substituent is called **phenyl (Ph)**



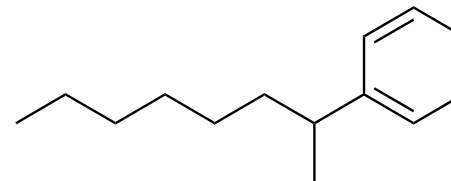
chlorobenzene



nitrobenzene

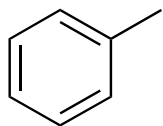


ethylbenzene

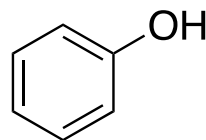


2-phenyloctane

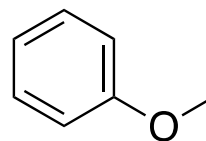
- Some substituted benzenes have important common names



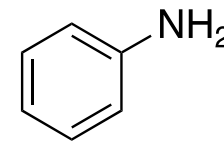
toluene



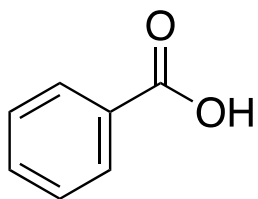
phenol



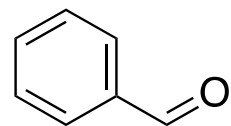
anisole



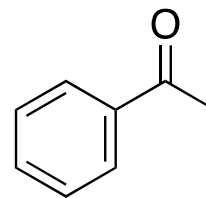
aniline



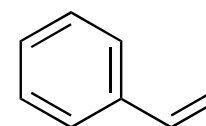
benzoic acid



benzaldehyde



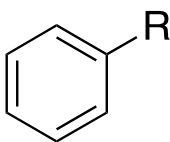
acetophenone



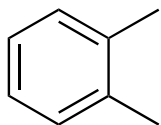
styrene

17.2 Disubstituted Derivatives of Benzene

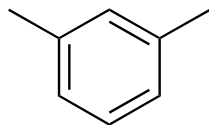
- The terms **ortho**, **meta**, and **para** are used to designate the carbons of a benzene ring in relation to a substituent *or* the relative positions of substituents with respect to each other
 - **Ortho**: 1,2 relationship
 - **Meta**: 1,3 relationship
 - **Para**: 1,4 relationship
- Positions with respect to a substituent R



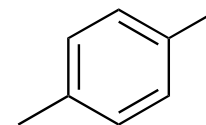
- Relations between two substituents



ortho-xylene



meta-xylene



para-xylene

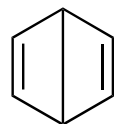
- Be aware of locant numbers in polysubstituted benzenes but you are not responsible for naming them

17.3 Proposed Structures of Benzene (Just a *Little* History...)

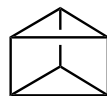
- The structure of benzene was hotly debated early in the history of organic chemistry
- It was known that benzene gives substitution products in reaction with electrophiles but reacts sluggishly and that it has the molecular formula C_6H_6



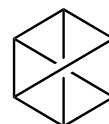
Hückel



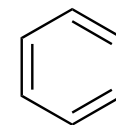
Dewar



Ladenburg

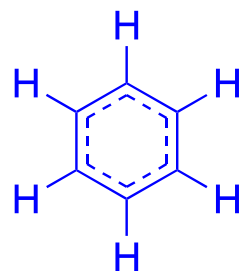


Klaus

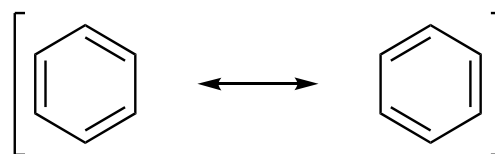


Kekulé

- Kekulé believed *incorrectly* that the two forms of his structure were rapidly interconverting isomers
- We now know that the two forms are resonance structures of a single resonance hybrid



resonance
hybrid

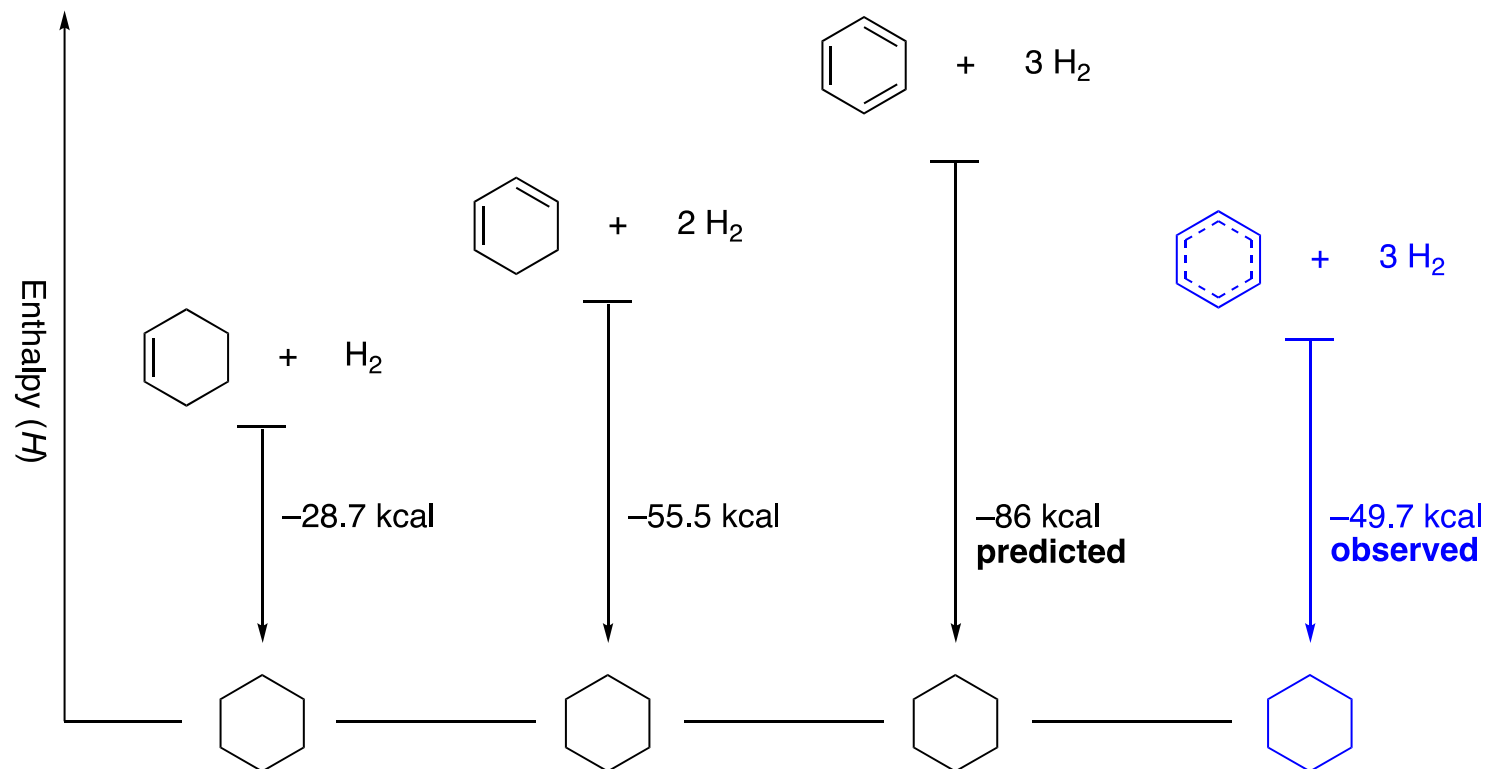


17.4 Benzene is Unusually Stable for a “Polyene”

- Benzene does not undergo electrophilic addition reactions like simple alkenes

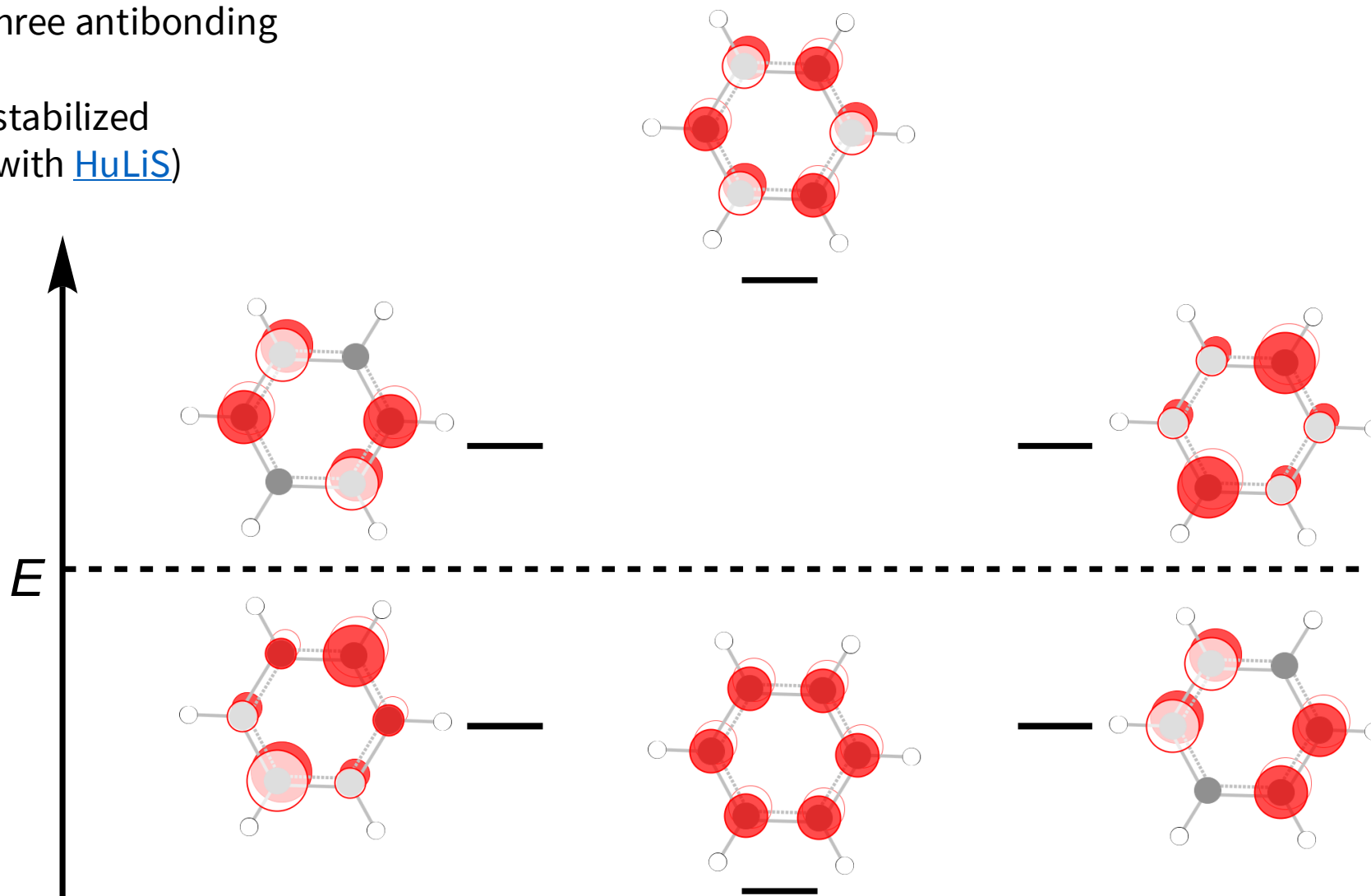


- Heats of hydrogenation suggest an **aromatic stabilization energy** of 36.3 kcal/mol for benzene



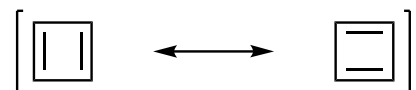
17.4 Pi Molecular Orbitals of Benzene

- The molecular orbitals of benzene look very different from those of an acyclic conjugated polyene
- Six MOs: three bonding and three antibonding
- Pi electrons are significantly stabilized in the cyclic structure (verify with [HuLiS](#))

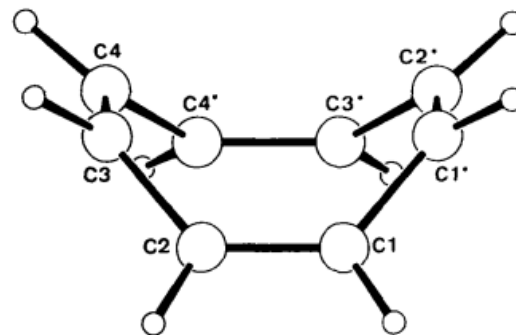
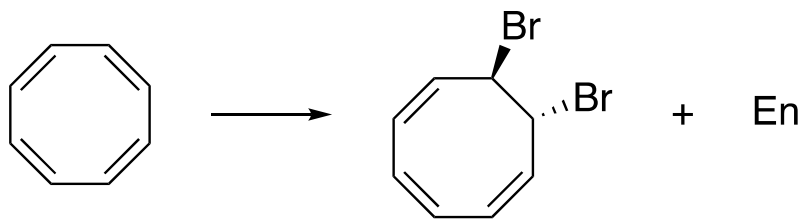


17.4 What is Aromaticity?

- There is more to aromaticity than just alternating single and double bonds...
- Cyclobutadiene has never been isolated; only observed under extreme conditions



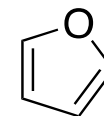
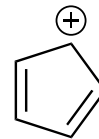
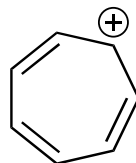
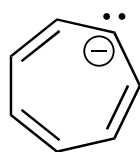
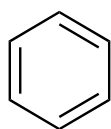
- Cyclooctatetraene readily reacts with bromine in an electrophilic addition reaction



- What makes a given structure aromatic?

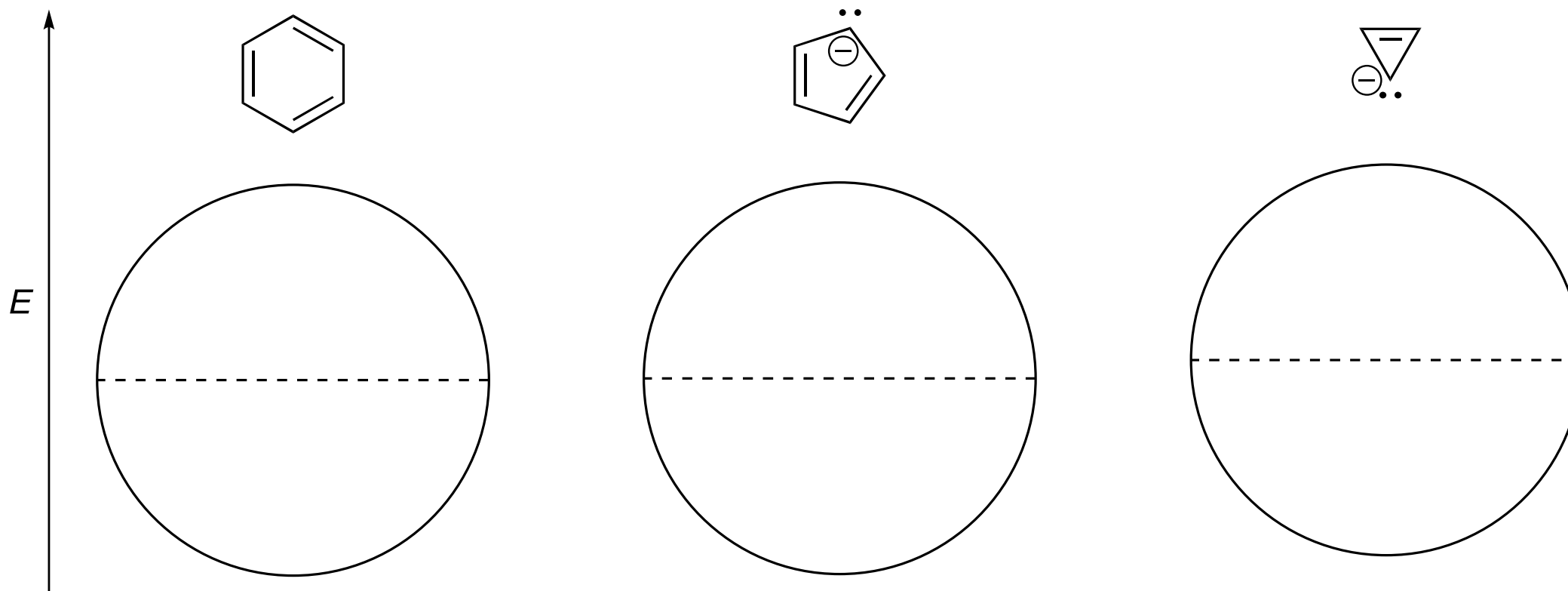
17.4 Criteria for Aromaticity

- For a molecule to display aromatic stabilization, four conditions must be met
 - **Cyclic.** Acyclic molecules cannot be aromatic
 - **Fully conjugated.** Each atom in the ring must have sp^2 or sp hybridization, enabling an unhybridized p orbital to participate in pi bonding
 - **Planar.** The ring must be planar to ensure good overlap of adjacent p orbitals
 - **Hückel's rule.** The number of pi electrons in the ring must be “two plus a multiple of four,” $4n + 2$
- “Anti-Hückel” molecules that satisfy the first three conditions but have $4n$ pi electrons are especially unstable and are called **antiaromatic**



- The energy levels of cyclic, planar, fully conjugated hydrocarbons follow a pattern
- Let's examine with [HuLiS...](#)

- Draw a circle and inscribe the structure in the circle with a vertex down
- A pi molecular orbital is located at each vertex
- Fill with the appropriate number of pi electrons



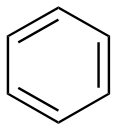
- $4n + 2$ pi electrons correspond to a fully filled set of bonding MOs
- $4n$ pi electrons correspond to either a diradical structure or one containing occupied antibonding MOs, an antiaromatic situation

17.5 Annulenes and Polycyclic Aromatic Hydrocarbons

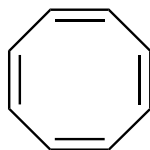
- Annulenes** are cyclic hydrocarbons containing alternating single and double bonds



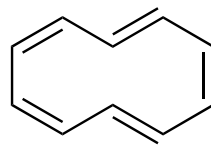
[4]annulene



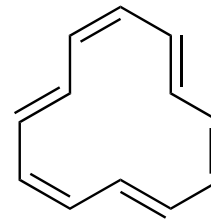
[6]annulene



[8]annulene

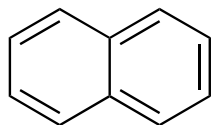


[10]annulene

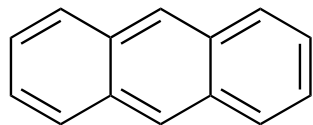


[12]annulene

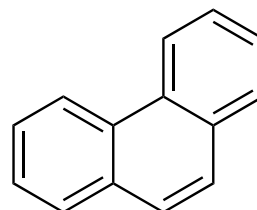
- Polycyclic aromatic hydrocarbons** consist of multiple unsaturated rings fused together



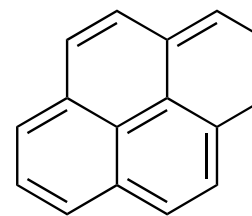
naphthalene



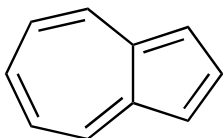
anthracene



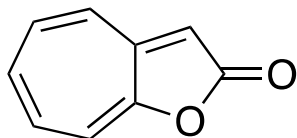
phenanthrene



pyrene

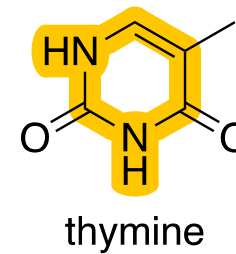
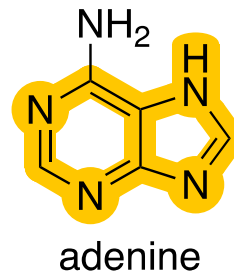
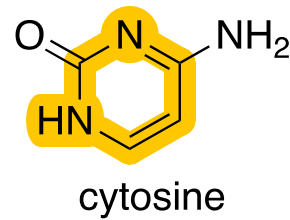
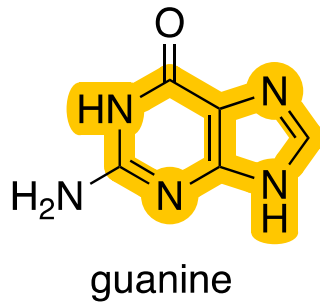
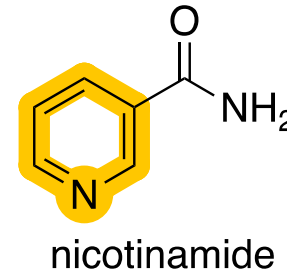
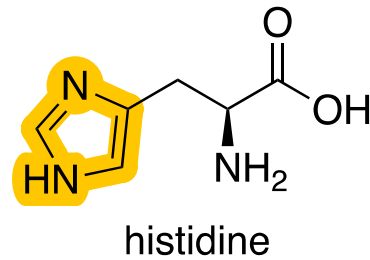
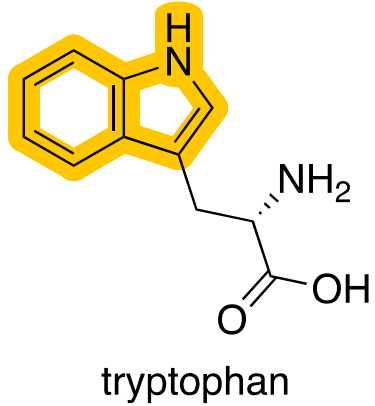


azulene

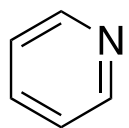


oxaazulanone

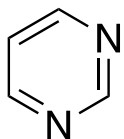
- Aromatic structures containing heteroatoms are known as **aromatic heterocycles** or **heteroarenes**
- Found in a wide variety of biomolecules, including amino acids and the nitrogenous bases of NAs



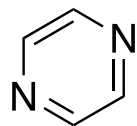
- What makes these compounds aromatic?
- What role do lone pairs play in aromatic pi systems?
- What can resonance reveal about aromatic heterocycles?
- How do reactions of aromatic heterocycles differ from those of related compounds?



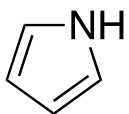
pyridine



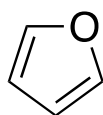
pyrimidine



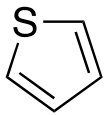
pyrazine



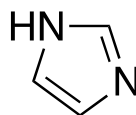
pyrrole



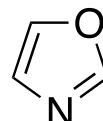
furan



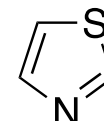
thiophene



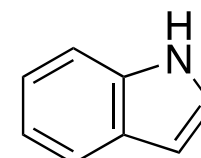
imidazole



oxazole



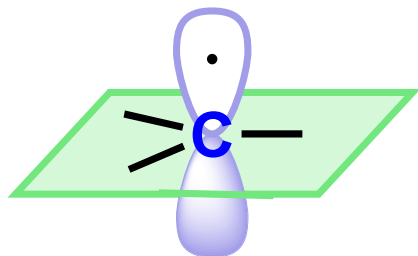
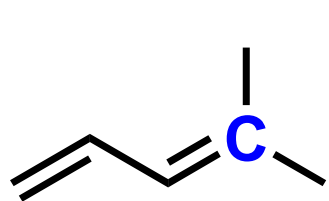
thiazole



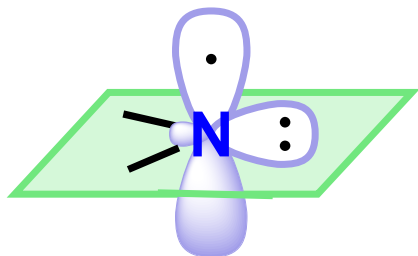
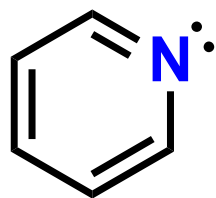
indole

17.5 Types of Atoms in Aromatic Heterocycles

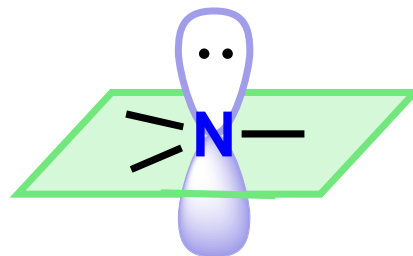
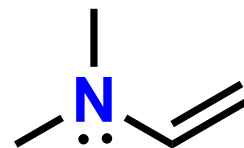
- Recall our earlier survey of heteroatoms in conjugated systems...it applies here!
- Heteroatoms that are *not* engaged in pi bonding contribute one and only one lone pair to the aromatic pi system



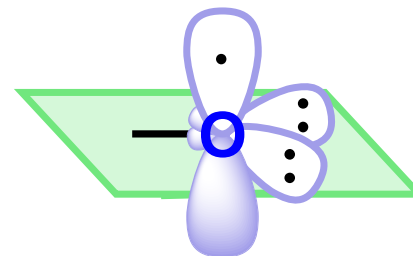
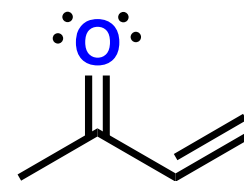
C



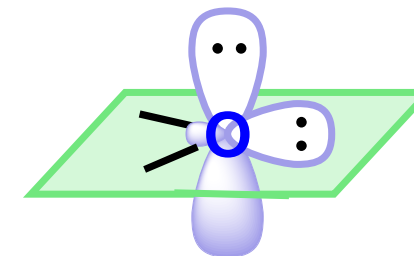
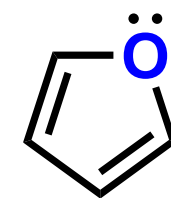
N2



N3

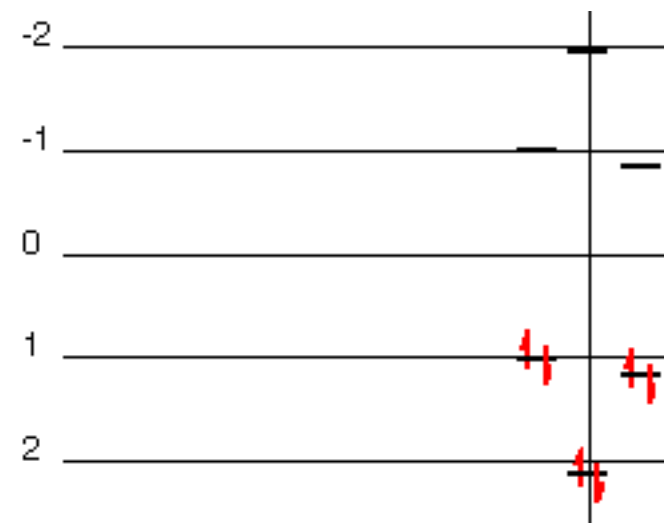
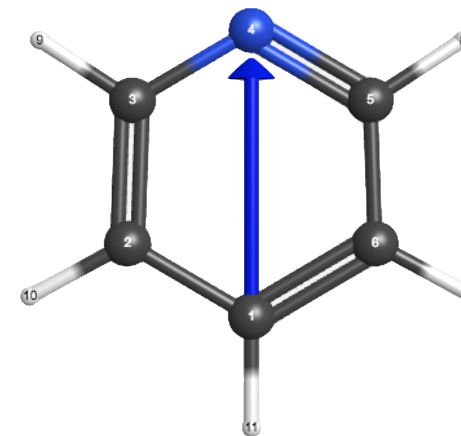
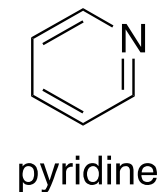


O1

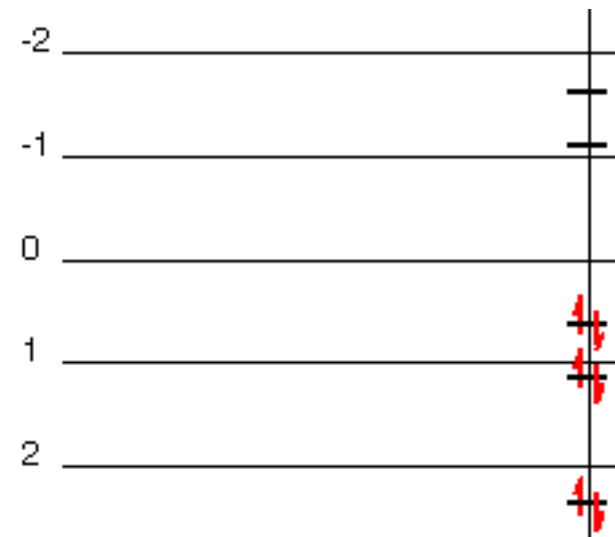
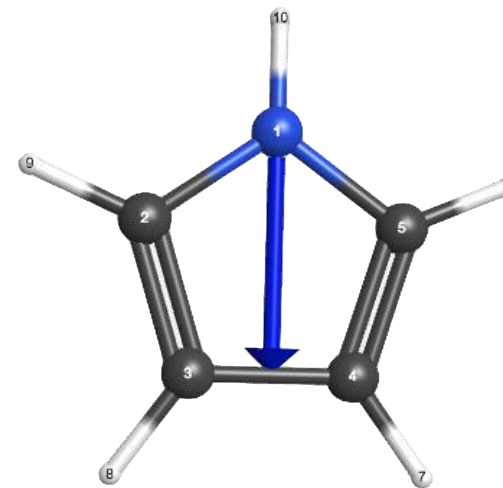
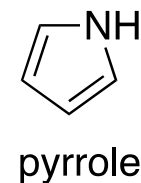


O2

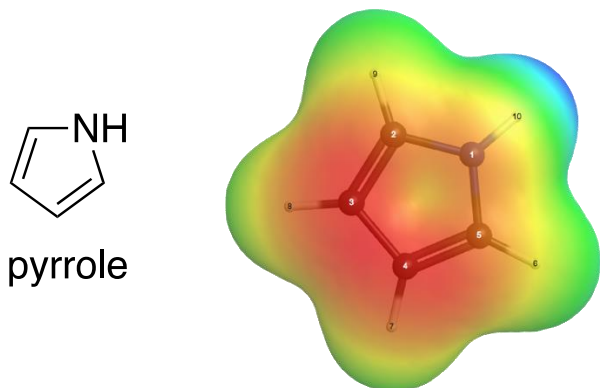
- Pyridine and pyrrole are fundamentally different aromatic heterocycles
- Pyridine** is an **electron-poor ring** containing an N2 nitrogen atom
- Resonance forms include **positive** charge on carbons in the ring
- The lone pair on nitrogen is *not* part of the aromatic pi system
- The molecular dipole vector points toward the nitrogen atom



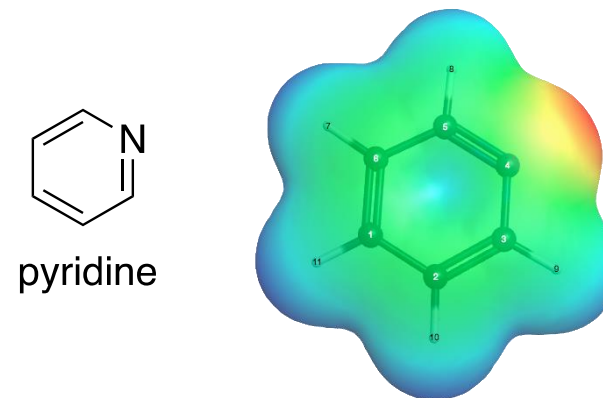
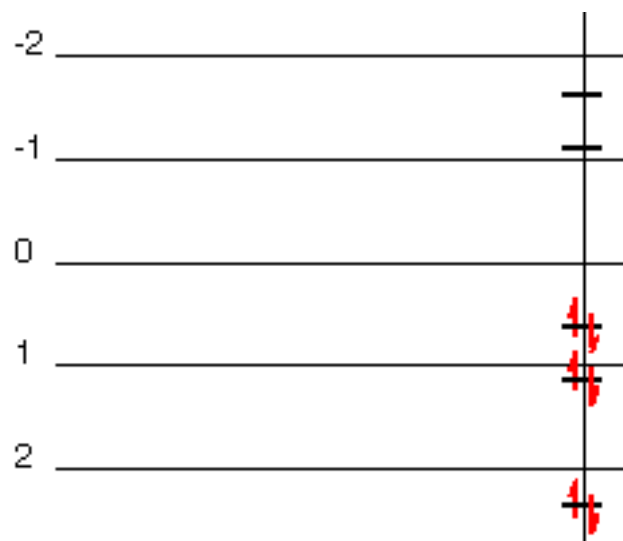
- Pyridine and pyrrole are fundamentally different aromatic heterocycles
- Pyrrole** is an **electron-rich ring** containing an N3 nitrogen atom
- Resonance forms include **negative** charge on carbons in the ring
- The lone pair on nitrogen *is* part of the aromatic pi system
- The molecular dipole vector points toward the carbon atoms



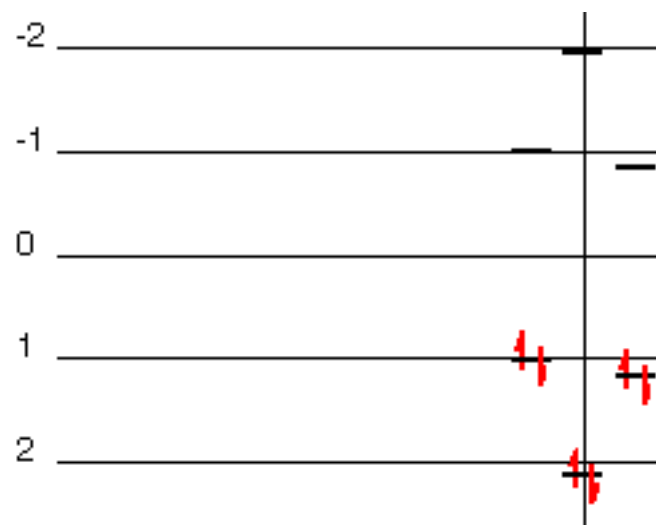
- Compare the orbital energies and electrostatic potential maps of pyridine and pyrrole



Pyrrole is an electron-rich ring.

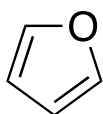


Pyridine is an electron-poor ring.

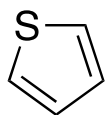


17.5 Other Examples of Aromatic Heterocycles | 1 of 2

- Aromatic heterocycles most often contain nitrogen, oxygen, or sulfur atoms and may be fused with benzene rings or other heteroaromatic rings

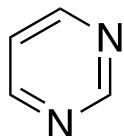


furan

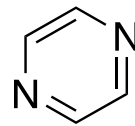


thiophene

Five-membered rings

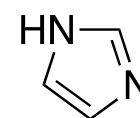


pyrimidine

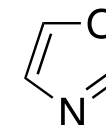


pyrazine

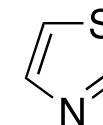
Six-membered azines



imidazole

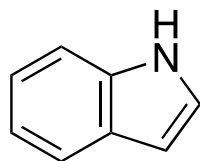


oxazole

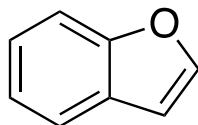


thiazole

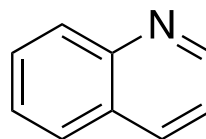
Azoles



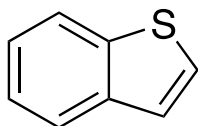
indole



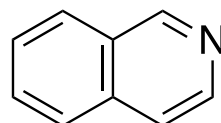
benzofuran



quinoline



benzothiophene



isoquinoline

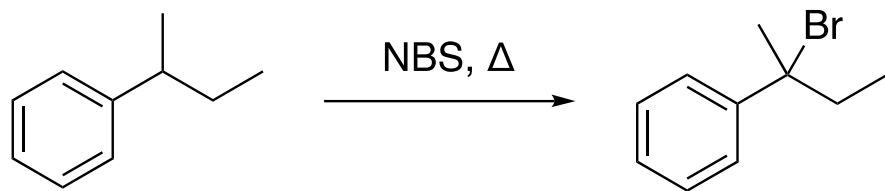
Fused heterocycles

- Aromatic stabilization energies (ASE) tend to be lower for heteroarenes than benzene

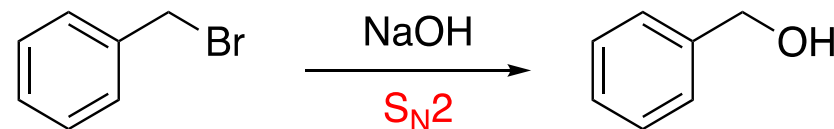
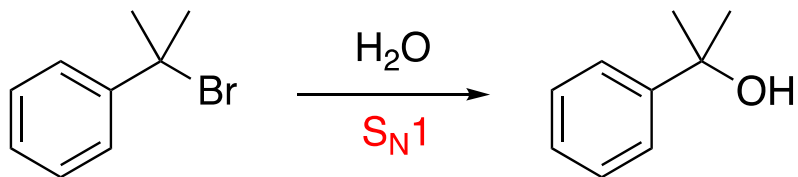
Compound	ASE (kcal/mol)
benzene	36
pyrrole	21
furan	15
thiophene	19
indole	73
quinoline	82
isoquinoline	83
pyridine	32
pyrimidine	33
pyrazine	32
pyridazine	26

- **Benzylic positions** are adjacent to aromatic rings
- Charges and radical character are stabilized at benzylic carbons by resonance (analogous to allylic positions)

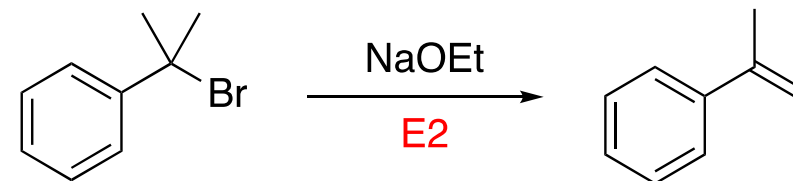
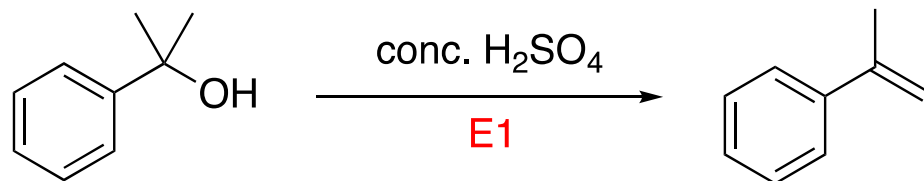
- Radical bromination



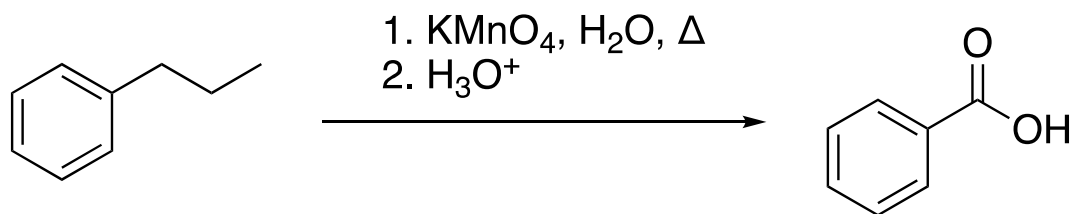
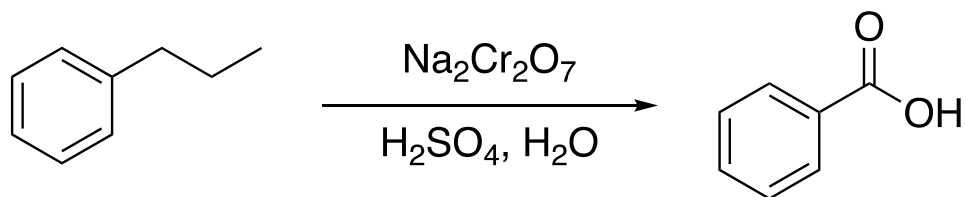
- Substitution reactions (S_N1 , S_N2)



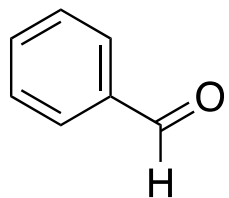
- Elimination reactions ($E1$, $E2$)



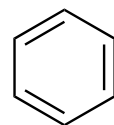
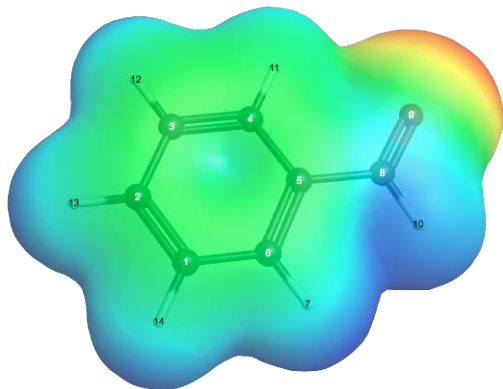
- Alkyl chains linked to a benzene ring can be oxidized to -COOH , provided the chain has at least one hydrogen linked to the benzylic carbon (**strong benzylic oxidation**)



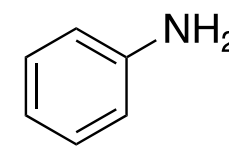
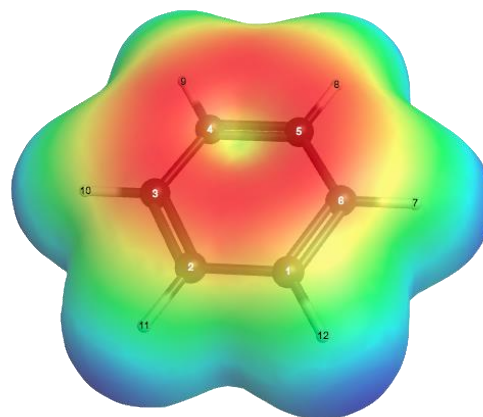
- Electron-donating and electron-withdrawing groups have a profound effect on electron density in benzenes
- These groups act via **resonance effects** (previously discussed) and **inductive effects**



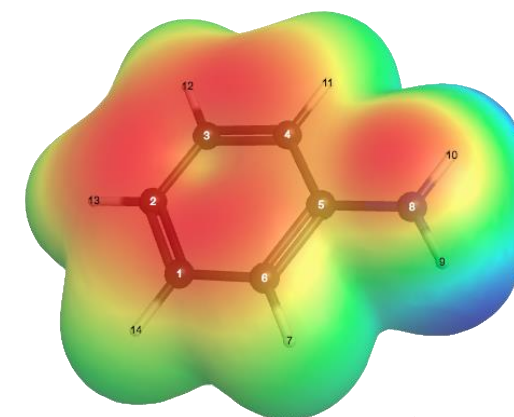
benzaldehyde



benzene



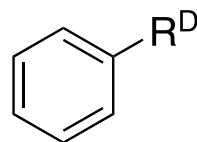
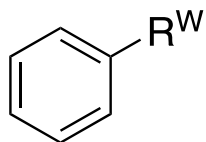
aniline



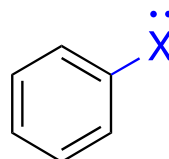
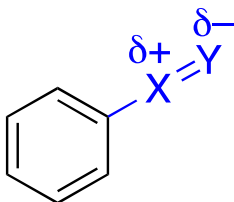
electron poor

electron rich

- **Inductive effects** involve the polarization of sigma bonds and can be understood using electronegativity

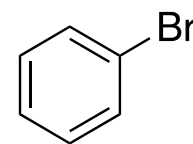
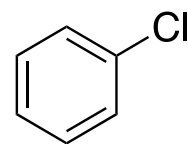
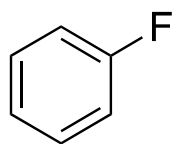
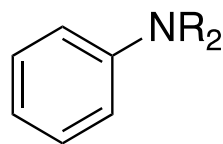
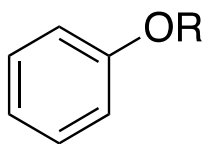


- **Resonance effects** involve delocalization of pi electrons and can be understood by “pushing electrons”

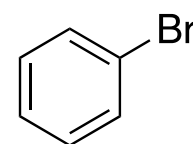
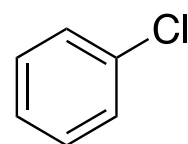
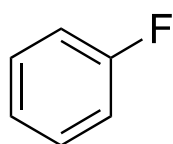
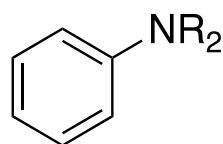
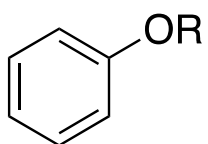


- In **electron-rich** benzenes, the ortho and para positions are especially **nucleophilic**
- In **electron-poor** benzenes, the ortho and para positions are especially **electrophilic**

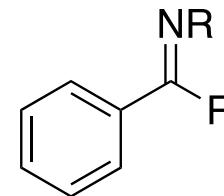
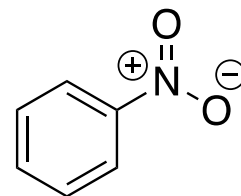
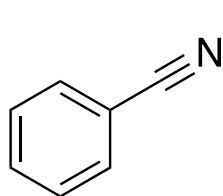
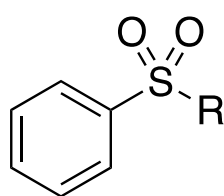
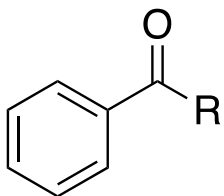
- Most heteroatomic substituents are withdrawing by induction (they're more electronegative than carbon)



- Alkyl groups are donating by induction
- Groups that are donating by resonance include $-NR_2$, $-OR$, $-X$ (halogens), $-SR$

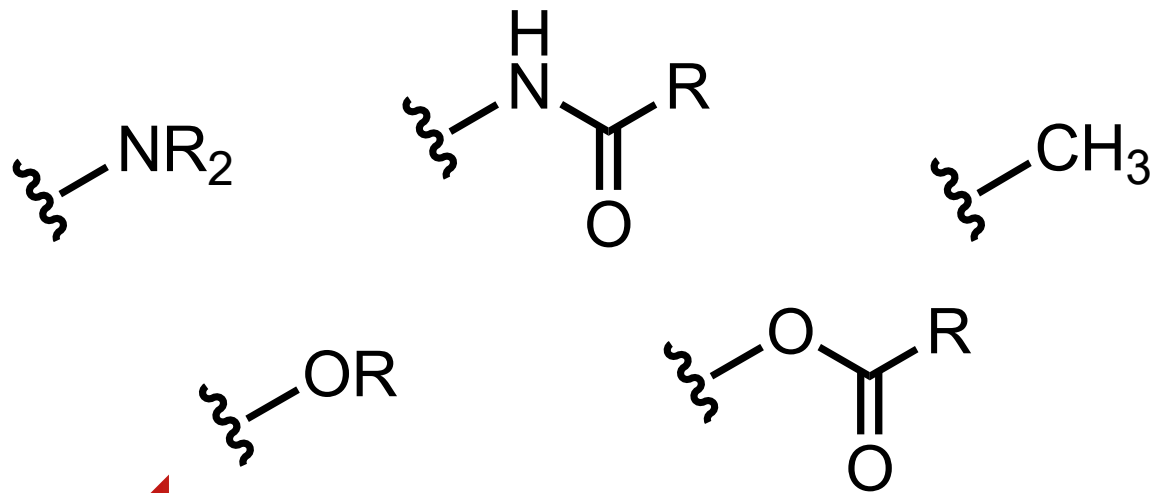


- Groups that are withdrawing by resonance include $-COR$, $-SO_2R$, $-CN$, $-NO_2$, $-C(=NR)R$

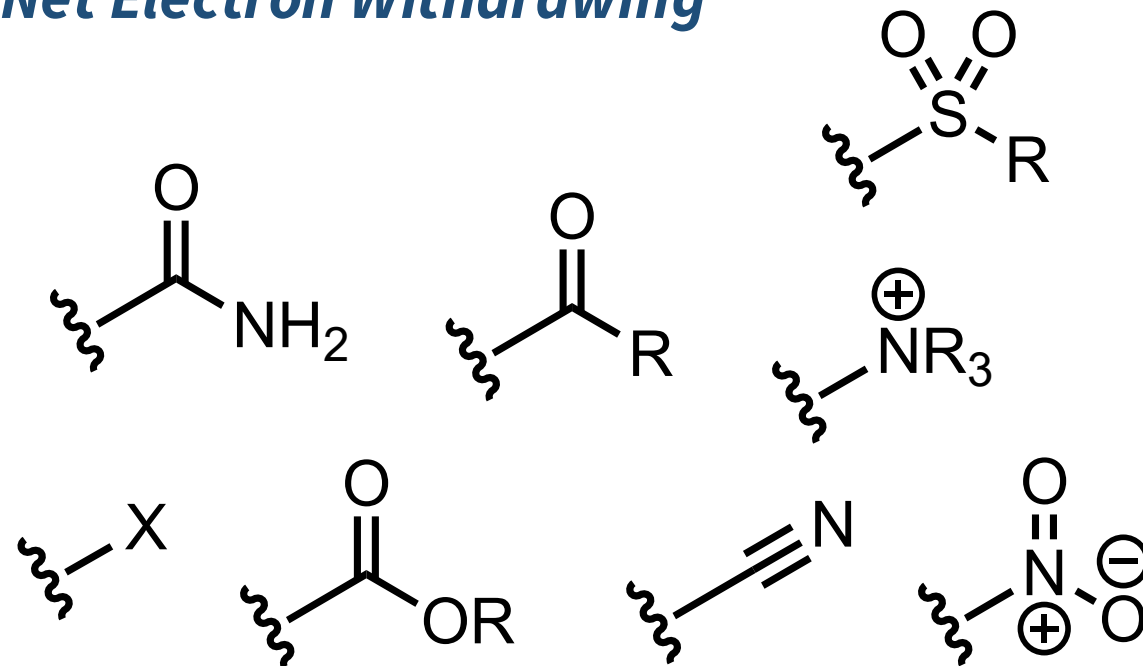


- Resonance effects generally outweigh inductive effects *except* for the halogens (large and electronegative)

Net Electron Donating



Net Electron Withdrawing



electron rich

electron poor